ECON 4660 / 8666 — Problem Set 3

International Economic Development · Fall 2026 · Prof. Zhigang Feng
Due Monday, November 23, before class (Canvas). At most 3 pages.

This set is *after* the midterm. It covers overlapping generations, social security, and remaining population/education ideas.

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1. Overlapping generations and pay-as-you-go social security

Each generation lives two periods. Population: $N'/N = 1 + n$ with $n = 0.1$. The consumer chooses $c = c'$ whenever possible. Income when young: $y = 150$. Income when old: $y' = 0$. Interest rate $r = 0$. A PAYG system taxes the young at rate t and pays a benefit b to each old person. No other taxes or spending.

(a) Optimal consumption as a function of t and b.

(b) The government levies a 20% social security tax on the young. Find b that balances the social security budget. (The number of old people is not equal to the number of young people.)

(c) Lifetime wealth, optimal consumption, and saving under that system.

(d) The same objects with no social security.

(e) Explain the effect of social security on welfare, consumption, and saving in this model.

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2. Concepts

(a) Limitation of the Malthusian model.

(b) Implicit vs. explicit cost of children.

(c) [If not already used on PS2 in your notes] How policies can affect long-run growth through human-capital accumulation.